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Learning 101 Reloaded: *Revisiting the Basics for the GenAI Era*

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***Abstract*—In this paper, we delve into the transformative landscape of education amidst the disruptive advances of generative AI (GenAI), characterized by an unprecedented capacity to generate new information with tools such as ChatGPT. Building upon my previous publication, "Learning 101: the Untaught Basics," in this magazine, I interrogate what has changed since then and how learners can harness the power of GenAI while avoiding its pitfalls, and learn how to learn effectively using GenAI tools. The dual role of GenAI in education is highlighted with immense potential for personalized learning side by side with the potent challenges it poses such as ethical concerns, intellectual property issues, biases, misinformation, and halluncations. Central to our argument is the development of metacognitive skills and discernment, enabling learners to navigate abundant resources effectively while avoiding overdependence and outsourcing of thinking to technology. This paper proposes strategies for integrating these tools into learning processes, aiming to equip educators and students with the skills to critically assess information and adapt to the evolving educational landscape. Our goal is to provide actionable insights for thriving in this new era, balancing the enhancement of learning and wisdom with technological advancements.**

I. Introduction

Welcome to the Generative AI (GenAI) era—a transformative time in education where answers to our questions are merely a prompt away. Picture this: you ask a question—any question—and within moments, you receive a coherent and accurate answer. This is the prowess of GenAI, a tool that simplifies the intricacies of subjects like algebra and history, providing insights with unprecedented ease [1].

However, such remarkable capabilities come with potential drawbacks. GenAI's ability to quickly furnish knowledge could inadvertently lead to an overreliance on technological solutions. If answers are just a query away, the incentive for students to engage deeply with learning materials might diminish. This phenomenon, known as "deskilling," threatens to erode our innate abilities for critical thinking and problem-solving, as we lean more on AI for answers [2].

This modern educational quandary requires careful navigation. GenAI's utility is undeniable, and its integration into our learning environments is here to stay. Yet, it is not a substitute for the rigorous, time-tested processes that underpin learning and intellectual development. Our challenge is to leverage GenAI judiciously, ensuring it serves as an adjunct to enhance learning, without compromising the development of our cognitive faculties.

Our previous research paper titled Learning 101 [3] critically examined commonly adopted intuitive learning approaches, revealing their misalignment with scientifically supported learning strategies. This mismatch is increasingly crucial to address in the GenAI era, where easy access to information might foster a deceptive sense of knowledge, overshadowing the need for deep, meaningful learning.

In this paper, we explore the critical juncture between established learning methodologies and the transformative capabilities of GenAI. Our objective is to empower learners to fully leverage these technological advancements while anchoring them in time-tested principles of effective learning. This includes fostering a learner's capacity for critical thinking beyond mere superficial engagement with information, enabling deep analysis, and facilitating the application and transfer of knowledge to novel contexts.

This dialogue gains urgency with the rise of GenAI technologies like ChatGPT, enhancing learning with conversational responses to various queries. However, while these tools promise personalized learning and access to tailored content, they also risk encouraging learners to sidestep traditional educational rigor by outsourcing critical thinking to AI.

Building on my earlier contributions to this magazine, "*Seven Learning Impediments and Their Remedies*" [4] and "*Learning 101: The Untaught Basics*" [3], this paper revisits essential learning principles in light of the GenAI era. It explores the complex relationship between human intelligence and artificial intelligence in achieving deep understanding and wisdom. Our goal is to equip learners for navigating not only today's technological realm but also the dynamic future of education shaped by GenAI.

II. GenAI Era Learning Impediments & Remedies

In the GenAI era, education extends into digital and AI-driven realms like GenAI [2], facing the dual challenge of harnessing technology's benefits while avoiding over-reliance that could undermine deep engagement in learning. Building on our previous exploration of seven learning impediments and their remedies [4], this section revisits and evaluates their relevance in the current context. We aim to revise and highlight how these impediments and their remedies adapt to the GenAI landscape, providing an updated perspective summarized in Table I.

1) *Lack of Metacognitive Skills*: Our previous study identified the deficiency in metacognitive skills—such as self-reflection and an understanding of one's own learning process—as a significant impediment to effective learning. This challenge remains highly relevant in the GenAI era. Remedying this involves enhancing learners' self-awareness and regulatory skills through strategies like self-reflection, active learning planning, and monitoring. GenAI tools can effectively support this development by providing personalized feedback, reinforcing the significance of metacognitive skills in leveraging technology for learning.
2) *Undue Emphasis on Speed and Fluency*: In the GenAI era, the instantaneous and plausible responses provided by AI tools can push learners towards prioritizing speed and fluency over deep understanding. With GenAI capable of generating quick outputs for any query, the temptation to bypass in-depth engagement with the subject matter increases. To counter this, it's important to promote depth and exploratory learning that extends beyond initial, surface-level responses. Encouraging learners to engage deeply with content, fostering critical evaluation of AI-generated information, and emphasizing the importance of comprehending processes over simply obtaining results are crucial strategies in this context.
3) *Underestimating Mistakes and Challenges*: Our previous study identified the neglect of the importance of mistakes and challenges as a significant impediment to learning. In the GenAI era, this issue has become even more pronounced. The seamless and effortless learning experiences offered by GenAI can inadvertently obstruct deep engagement and subject mastery. Acknowledging the indispensable role of struggles and errors is crucial for effective learning. It may be necessary to intentionally restrict GenAI access during certain learning stages, ensuring learners encounter the vital challenges and mistakes that are fundamental to achieving a comprehensive understanding.
4) *Inadequate Critical Thinking Skills*: In an era where AI can often provide misleading or incorrect information, the reliance on these tools can hinder the development of critical thinking skills. It's crucial to understand that the use of AI technology does not eliminate the need for critical analysis and independent thought. Enhancing critical thinking skills alongside AI becomes imperative. This includes promoting manual problem-solving and critical assessment of AI-generated solutions before relying on them. Encouraging learners to question and creatively engage with AI tools, rather than accepting their outputs unquestioningly, is essential for nurturing originality and a discerning mindset.
5) *Insufficiently Diverse Learning Experiences*: A well-rounded education necessitates a rich variety of learning experiences, often missing in current learning environments. To address this, it's essential to cultivate a broad, multimodal learning spectrum. This involves leveraging GenAI to access and engage with a wide range of subjects and learning styles. Simultaneously, it's important to promote hands-on and experiential learning activities, blending these traditional approaches with the innovative capabilities of AI. Integrating diverse methodologies enhances the depth and breadth of the learning experience.
6) *Overlooking Social Learning Aspects*: Learning is deeply *situated* in social environments, i.e., learning is not just an individual process but occurs within and is shaped by social interactions and contexts. To address this, it's essential to integrate social aspects into learning frameworks. This can be achieved by blending AI with collaborative projects and discussions that foster meaningful social interactions. Such an approach should not aim to replace, but rather to complement and enhance, interactions with peers and mentors. It emphasizes the importance of community engagement and practical group tasks, recognizing that knowledge and skills are often best developed and understood within the context of their social and practical applications.
7) *Neglecting Lifelong Learning and Wisdom*: There's a tendency to focus on immediate learning goals, overlooking the importance of continual education and

TABLE I: GenAI Era Learning Impediments & Remedies (Adapted from Qadir, 2015 [4])

***Impediment 1*: Lack of Metacognitive Skills** *Solution: Enhance self-regulated learning skills* - Foster self-reflection and comprehension evaluation strategies - Promote active learning planning and monitoring - Utilize GenAI for feedback in self-regulated learning
***Impediment 2*: Overemphasis on Speed and Fluency** *Solution: Promote depth over superficial understanding* - Encourage deep learning beyond initial responses - Foster critical evaluation of AI-generated content - Emphasize process comprehension over result orientation
***Impediment 3*: Underestimating Mistakes and Challenges** *Solution: Embrace learning potential in errors and difficulties* - Encourage growth mindset to view challenges as opportunities - Use GenAI scenarios for error exploration and learning - Promote resilience and persistence against setbacks
***Impediment 4*: Inadequate Critical Thinking Skills** *Solution: Enhance critical thinking alongside AI* - Promote manual problem-solving before AI reliance - Teach critical assessment of AI solutions - Encourage creativity and originality with AI assistance
***Impediment 5*: Insufficiently Diverse Learning Experiences** *Solution: Cultivate broad multimodal learning spectrum* - Use GenAI for diverse subjects and styles - Promote hands-on and experiential learning activities - Integrate traditional methods with AI-enhanced formats
***Impediment 6*: Overlooking Social Learning Aspects** *Solution: Foster social interaction in learning* - Blend AI with group projects and discussions - Supplement, not replace, peer and mentor interactions - Promote community engagement and group tasks
***Impediment 7*: Neglecting Lifelong Learning and Wisdom** *Solution: Encourage ongoing learning and wisdom cultivation* - Advocate for continuous education and adaptation - Emphasize human experience and insight alongside AI - Promote varied life experience reflection and learning

wisdom cultivation. Emphasizing ongoing learning and adapting to new knowledge is crucial. Highlighting the value of human experience and insight alongside AI, as well as promoting reflection and learning from diverse life experiences, are key strategies in fostering lifelong learning.

III. Navigating Paradoxes and Timeless Principles

"*The more things change, the more they remain the same*", an aphorism by Jean-Baptiste Alphonse Karr.

This work highlights the critical need for strong metacognitive foundations and adherence to learning fundamentals, especially when technology like GenAI tools can eclipse critical thinking and deep learning. In an era defined by instant access to answers via technologies such as ChatGPT, recognizing our knowledge gaps becomes vital, embodying Socratic wisdom—the awareness of the extent and limits of our understanding. The GenAI era amplifies learning paradoxes: while information is plentiful and easily accessible, it can both facilitate and complicate our pursuit of knowledge.

Now, we come to two terms, *undesirable fixations*" and *desirable difficulties*," introduced in my previous "Learning 101" paper [3]—terms that continue to remain highly relevant as we engage with the GenAI-driven educational landscape. These concepts illuminate the nuanced challenges and opportunities presented by GenAI in learning. *Undesirable fixations* on perfection, fluency, and discipline can be exacerbated by GenAI tools, risking a shallow approach to learning. *Desirable difficulties*, essential for deep and meaningful learning, offer a counterbalance, promoting a more nuanced engagement with technology. We discuss these two concepts and their relevance for GenAI-era education in more detail in turn next.

A. Undesirable Fixations in the GenAI Era

Our intuitive models of learning are often misaligned with the realities of effective learning, especially in an era dominated by GenAI technologies.

1) The False Idol of Perfection: In the GenAI era, the seemingly flawless knowledge provided by AI tools might mask the true essence of learning—a process fundamentally rooted in making and learning from mistakes. There's a danger that learners could adopt a fixed mindset, overly relying on AI for solutions, which may hinder their intellectual development. Recognizing the significance of making mistakes is key; they play a vital role in achieving deep understanding and retaining knowledge. While GenAI is a powerful aid in the learning journey, it cannot substitute the invaluable insights gained through navigating and rectifying errors.

2) The Illusion Created by Fluency: The ease with which GenAI facilitates learning can create a misleading sense of expertise. While these tools expedite knowledge acquisition, they cannot instill the nuanced understanding that comes with active participation, ongoing practice, and the application of concepts in diverse situations—abilities that surpass GenAI's scope. With the deluge of information available, learning now requires a more robust approach that includes critical assessment and synthesis—skills where GenAI can assist but not fully take over the intricate thinking needed for thorough understanding and retention.

3) Discipline and Variety in Learning with GenAI: The balance between personal effort and GenAI support is a critical element in contemporary education. As GenAI simplifies access to information, learners should periodically practice working analog where they have to exercise their innate unaided selves. This will require a healthy mixture of working in the real world and in the digital world with and without technological aids. Moreover, in the GenAI era, the interplay between focus and breadth in learning takes on new importance. A well-rounded education, one that equips learners for the rapid changes of our time, depends on a harmonious blend of concentrated learning and broad, varied exposure.

B. Desirable Difficulties in the GenAI Era

The concept of "*desirable difficulties*" [3] remains relevant in the GenAI era, emphasizing the importance of effortful learning and the futility of overreliance on technological aids.

1) Effortful Learning with GenAI: It is important to continue to leverage techniques such as retrieval practice (where learned information is recalled from memory) in addition to having answers generated by GenAI. This combination will lead to robust learning, in which learning is not completely outsourced to technology. Additionally, the concept of *GenAI as a facilitator, not a shortcut* is crucial in the modern learning context, implying the use of GenAI toolsas an aid to enhance the learning experience, not as a replacement for the intrinsic effort required in the learning process.

2) Importance of Spaced Learning in GenAI Era: The GenAI era presents a nuanced challenge to traditional learning. GenAI's quick access to information contrasts with time-tested learning that underpins durable memory. Swift answers from GenAI don't afford the cognitive depth needed for thorough knowledge absorption and processing. Therefore, the value of retrieval practice and spaced learning is heightened. These methods are key for lasting memory, and while GenAI can support them, it shouldn't override the need for deep internalization of knowledge. Effective GenAI integration in education requires a delicate balance with human cognitive rhythms, appreciating its benefits while recognizing its limitations.

3) Embracing Disfluency in the GenAI Era: In the GenAI era, the convenience of instant answers poses a paradox: the essential role of effort in learning may be overshadowed. It is important to remember that disfluency—which results from engaging with challenging material—continues to be important for deep learning. Mastery involves not just ease of access but grappling with complex concepts. A balance between cognitive exertion and digital facilitation is desirable, which promotes efficiency without the atrophying of natural capabilities. Embracing this disfluency, educators should craft experiences that extend beyond learners' comfort zones, promoting critical thinking and adaptability.

IV. Optimizing Learning in the GenAI Era: Embracing Effective Techniques

"*The goal of education is not to increase the amount of knowledge but to create the possibilities for a child to invent and discover, to create men who are capable of doing new things*."—Jean Piaget.

A. Three Laws of GenAI Era Learning

In the context of GenAI Era Learning, the paper introduces three guiding imperatives, depicted in Figure 1, which serve as foundational principles for ensuring a human-centered approach to educational technology.

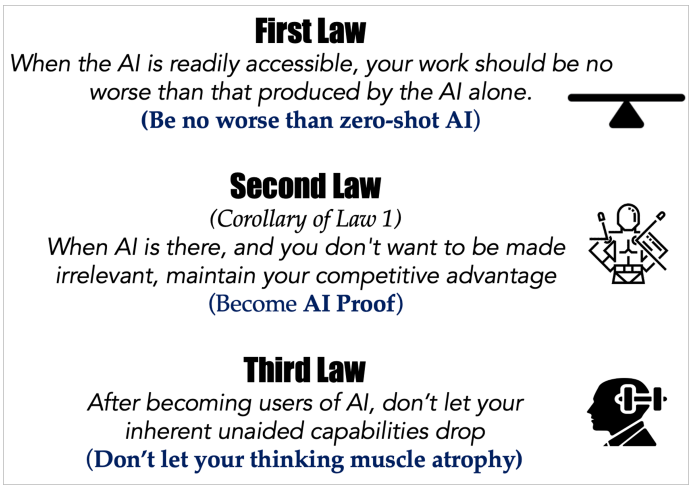


Fig. 1: Laws of Human-Centered AI-Augmented Education.

The *First Law*, "***Be no worse than zero-shot AI***," emphasizes that when AI tools are readily accessible for educational purposes, the outcomes from human effort should be at least as effective as those generated by the AI independently. This principle motivates educators and learners to go beyond merely matching the baseline of *zero-shot AI*, which refers to unoptimized AI output. Maintaining human competitive advantage is crucial, and this can be readily accomplished through AI literacy and effective human-AI partnerships.

The *Second Law*, "***Become AI-proof***," addresses the potential obsolescence of human roles in the face of advancing AI. It advises that to avoid becoming irrelevant, individuals must leverage their unique human qualities and foster skills and attributes such as creativity, critical thinking, and emotional intelligence, which AI cannot yet adequately replicate.

The *Third Law*, "***Don't let your thinking muscle atrophy***," warns against the potential cognitive decline that might occur if individuals become overly reliant on AI for thinking and problem-solving tasks. After integrating AI into learning processes, it is crucial to continue exercising and developing one's inherent abilities to prevent decay of thinking capability. This law suggests that while AI can augment learning, it should not replace the fundamental human processes of understanding, reasoning, and innovation.

Together, these laws form a framework for Human-Centered AI-Augmented Education, ensuring that AI serves as an enhancement rather than a replacement for human intellectual growth.

B. Learning Tips for the GenAI Era

Effective learning entails understanding human learning and memory; utilizing techniques for better retention; monitoring one's learning journey; and recognizing biases in learning judgments. In the GenAI era, it also involves understanding the role of GenAI tools and using these tools judiciously. Building upon insights from our previous "Learning 101" paper [3], let's explore how GenAI technology can enhance various learning techniques. These include:

1) *Spaced Learning with GenAI*: Expanding our understanding of spaced learning, which involves repeated exposure to information over increasing intervals. GenAI tools offer personalized review schedules, diverse content, and scenarios, optimizing long-term memory retention and fostering deeper learning.
2) *Interleaved Study Sessions with GenAI*: Exploring interleaved study sessions, where different topics are mixed during study sessions. GenAI tools provide diverse content and scenarios, enhancing comprehension and contextual understanding.
3) *Testing and Retrieval Practice with GenAI*: Let's dive into testing and retrieval practice techniques, which involve actively recalling information through quizzes or practice tests. GenAI facilitates immediate feedback, provides dynamic learning environments, and offers personalized experiences, enhancing mastery and reinforcing learning effectively.
4) *Collaborative Learning Enhanced by GenAI*: Finally, let's examine collaborative learning, emphasizing group discussions, problem-solving, and project-based learning. GenAI supports collaborative environments, nurturing social learning skills and collective comprehension of concepts.

Integrating these strategies into GenAI-enhanced education shifts the focus to a holistic, process-focused learning experience, actively engaging learners and fostering deeper understanding and skill development. This aligns with the evolving needs of modern education.

C. GenAI Era Student Primer for Engineering Education

In our discussion on evolving pedagogies for the GenAI era, we elaborate on strategies that prepare students for the intersection of AI and engineering education. Drawing from the insights of our previous paper [5], we propose a framework that goes beyond mere technical proficiency. Our framework, summarized in Table II, urges students to begin with a forward-looking mindset, setting goals that anticipate the deep integration of AI in future careers while balancing this with the need for human-centric skills. This dual focus is intended to prepare students for a future where AI augments human capability, without overshadowing the necessity for human discernment and innovation.

TABLE II: Adapting the Student Primer for Thriving in Engineering Education for GenAI Era
(*Adapted from Qadir & Al-Fuqaha, 2020 [5]*)

***Step 1*: Begin with the End in Mind for the Future**
A: Aim for careers enriched by deep AI knowledge. **B**: Plan careers with both AI and human skills in mind. **C**: Grow thinking skills with AI's advancements.
***Step 2*: Develop your Metacognitive Skills**
A: Deeply understand types and levels of learning **B**: Avoid common pitfalls (including AI-induced deskilling risks) **C**: Learning how to learn effectively in the GenAI Era
***Step 3*: Aim for Holistic and Reflective Learning**
A: Become a systems thinker **B**: Become a multi-model thinker leveraging AI as a tool **C**: Complement AI by being well-rounded and humane
***Step 4*: Learn from Feedback**
A: Pursue feedback (experts and AI) and become coachable **B**: Become a critical thinker and develop critical AI literacy **C**: Develop self-assessment and corroboration skills
***Step 5*: Take Ownership of Your Learning Journey**
A: Be self-directed and engage mindfully with AI Tools **B**: Enhance, don't replace, your learning with AI **C**: Collaborate effectively with other humans and AI
***Step 6*: Develop Authentic Skills with a Focus on Transferability**
A: Focus on your education and not just on obtaining answers **B**: Learn as you would in the real world without artificial constraints **C**: Use dialogic interactions with humans and AI to gain depth
***Step 7*: Embrace Lifelong Learning**
A: Be committed to lifelong iterative learning **B**: Master instrumental knowledge (including critical AI literacy) **C**: Practice and refine key lifelong learning skills for the GenAI Era

The framework presented in Table II offers a structured approach to learning in the GenAI Era, emphasizing the importance of both human and artificial intelligence in shaping a robust educational journey. It starts with a forward-looking mindset, encouraging students to envision their future careers in a world deeply integrated with AI, focusing on how AI can enhance their critical thinking abilities.

In their educational journey, learners must hone metacognitive skills to discern various learning strategies and proactively guard against AI-induced skill erosion. This self-aware approach lays the groundwork for holistic and reflective learning, urging students to adopt a systems-thinking perspective, utilize AI as a supportive tool, and remain well-rounded and in touch with the human aspects of their education.

Ownership of learning is paramount. Students are encouraged to direct their educational paths proactively, using AI tools as aids—not replacements—in their quest for knowledge. This self-driven approach is complemented by seeking feedback from both experts and AI systems, cultivating coachability, critical thinking, and the ability to assess one's progress.

The development of authentic skills, which are transferable to real-world situations, is highlighted, stressing the importance of learning beyond just finding answers and engaging in meaningful dialogues with both humans and AI. This transferability ensures that skills remain relevant and adaptable across various scenarios.

The framework prioritizes lifelong learning, urging a dynamic commitment to iterative education informed by AI and ethics. It readies students for continuous improvement, fostering resilience and adaptability for technological advancement. The goal is to bolster critical thinking and competitive edge in the GenAI Era.

V. Conclusions

In the Generative AI (GenAI) era, effective learning requires a nuanced approach that blends traditional methods with the capabilities of GenAI tools. To learn optimally, one must engage with enthusiasm and invest genuine effort, not just in absorbing information but in critically evaluating and integrating it. It's crucial to recognize and resist the illusions of learning, such as mistaking the ease of access to information for true understanding or confusing the fluency provided by AI tools with mastery. Optimal learning techniques, including retrieval tests, should be complemented with the strategic use of GenAI tools to enhance understanding, encourage deeper exploration, and facilitate the application of knowledge in varied contexts. This balanced approach ensures a robust and enduring mastery of subjects in an age where technology and human intellect intersect.

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References

[1] J. Qadir, "Engineering education in the era of ChatGPT: Promise and pitfalls of generative AI for education," in *2023 IEEE Global Engineering Education Conference (EDUCON)*. IEEE, 2023, pp. 1–9.

[2] A. Hamilton, D. Wiliam, and J. Hattie, "The future of AI in education: 13 things we can do to minimize the damage," Aug 2023. [Online]. Available: https://osf.io/preprints/edarxiv/372vr

[3] J. Qadir and M. A. Imran, "Learning 101: The untaught basics," *IEEE Potentials*, vol. 37, no. 3, pp. 33–38, 2018.

[4] J. Qadir, "What every student should know: Seven learning impediments and their remedies," *Potentials, IEEE*, vol. 34, no. 3, pp. 30–35, 2015.

[5] J. Qadir and A. Al-Fuqaha, "A student primer on how to thrive in engineering education during and beyond COVID-19," *Education Sciences*, vol. 10, no. 9, p. 236, 2020.